

HippoCortex

Biologically-Inspired Continual Learning for Edge Robotics
via Sharp Wave Ripple Generative Replay in State Space Models

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PROBLEM SCOPE

Continual learning is the problem of training a model on a sequential stream of tasks without erasing what it already knows [2]. Standard neural networks cannot do this. Each new task overwrites previously acquired knowledge, a failure mode called *catastrophic forgetting* [1]. The standard workaround, experience replay [3], keeps a buffer of raw past samples and retrains on them periodically. While effective on static benchmarks, this approach is computationally prohibitive for deployed robotics.

Raw sensory buffers grow with every new environment. Storing camera frames, LiDAR sweeps, and depth maps from every task a robot has ever encountered is neither memory-efficient nor privacy-safe. On an edge device like the Unitree Go2 Edu, encountering new terrains, new objects, and new people across its deployment lifetime, replay-based methods simply do not scale.

The Go2 Edu needs to adapt continuously: recognise a new terrain without forgetting the previous one, and learn a new object interaction without losing grip on past experience. Current continual learning methods struggle to simultaneously deliver: **zero raw data storage**, **linear-time architecture**, and **real-world embodied deployment**.

BIOLOGICAL INSPIRATION

The mammalian brain solved catastrophic forgetting long before machine learning existed. It runs two memory systems at different timescales: the **hippocampus** rapidly encodes new experience into temporary storage during waking hours, and the **neocortex** slowly integrates knowledge into stable long-term representations [4]. During sleep, the hippocampus does not replay raw sensory data. It fires brief high-frequency bursts (*sharp wave ripples*, SWRs) that transmit *compressed* memory traces to the neocortex for consolidation. No raw storage. Constant memory footprint. As illustrated in Figure 1, HippoCortex maps this mechanism directly onto a machine learning framework.

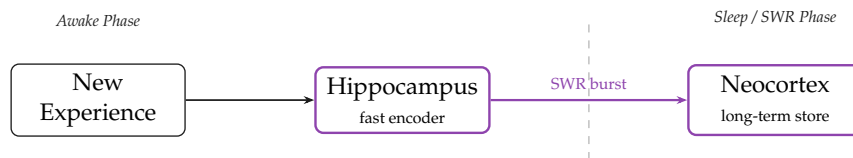


Figure 1: Biological memory consolidation mapped to the HippoCortex framework.

PROPOSED SOLUTION

The HippoCortex architecture translates the biological two-system memory model into a trainable framework. Rather than buffering raw data, it will compress task experience into compact statistics and regenerate synthetic memory traces during consolidation, mirroring what the sleeping brain does via SWRs. The project will run in two stages: Stage 1 will build and validate the core algorithm; Stage 2 will deploy it on a physical quadruped robot.

Stage 1: Core Algorithm (Months 1–3)

What is a State Space Model? State Space Models (SSMs) are neural networks that process sequential data through a recurrently updated hidden state, achieving *linear-time* computation. The Mamba architecture [5] extends this with selective state updates, matching Transformer accuracy at a fraction of the memory cost, making it well-suited for edge deployment.

HippoCortex will be built on this **Mamba SSM** backbone, improving upon recent SSM-based continual learning frameworks such as Mamba-CL [8] and Inf-SSM [9]. After each task, a compact statistics buffer will store only the mean and variance (μ, σ^2) of the Mamba hidden states, never raw inputs. A **conditional VAE** [6] (the SWR Generator) will sample synthetic past hidden states from this buffer during consolidation, replicating hippocampal SWR bursts. As shown in Figure 2, these synthetic states will pass through a **Null-Space Gradient Projector** [7], $P = I - UU^\top$ (where U spans the feature subspace of prior tasks), that will deflect new-task gradients away from previously learned representations. New learning will not overwrite old knowledge. The memory footprint will remain **constant**, regardless of tasks seen.

Validation will be conducted on Split-CIFAR100 (20 tasks) and ImageNet-R. Target: $\sim 8\text{--}10\%$ average accuracy gain over Mamba-CL [8]; $\sim 4\%$ over Inf-SSM [9].

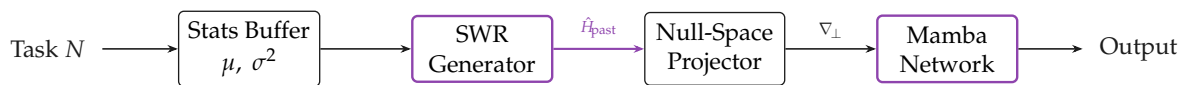


Figure 2: HippoCortex architecture during the SWR generative replay consolidation phase.

Stage 2: Embodied Deployment (Months 4–10)

Benchmark accuracy confirms an algorithm works in simulation; it does not confirm that it works on a physical agent facing sensor noise, compute constraints, and real task variation. Stage 2 will move HippoCortex from a classification dataset to a deployed robot.

HippoCortex will be extended into a **hybrid Mamba + Transformer architecture**. Mamba layers will handle efficient temporal state tracking across the robot’s sensor streams; selective Transformer attention layers will extract rich spatial features from the RGB camera and RealSense D435i depth frames. The HippoCortex SWR module will wrap the full hybrid stack (Figure 3); continual learning will apply across both layer types without architectural modification.

Inference will run entirely on the **Jetson Orin NX 16GB** (100 TOPS) with no remote base station. The sequential task stream on the Go2 Edu will be structured as: terrain recognition → object interaction → human proximity awareness → new environment mapping. Each will be learned in turn, without forgetting previous ones.



Figure 3: Information flow in the Stage 2 hybrid vision architecture.

EXPECTED OUTCOMES

Stage	Deliverable
Stage 1	$\sim 8\text{--}10\%$ accuracy gain over Mamba-CL; $\sim 4\%$ over Inf-SSM on Split-CIFAR100
Stage 1	Peer-reviewed publication: HippoCortex SWR replay on Mamba SSM
Stage 2	Hybrid Mamba-Transformer continual learning architecture
Stage 2	Deploy continual learning model to Go2 Edu
Stage 2	Second peer-reviewed publication on embodied continual learning

HARDWARE PLATFORM: UNITREE GO2 EDU

An embodied platform grounds research claims in the physical world: real sensor noise, real power limits, and real spatial variation that no simulation fully reproduces. Stage 1 requires no robot. Stage 2 requires a platform combining sufficient on-board computation with a rich multi-modal sensor stack. The Go2 Edu is the only consumer quadruped that delivers both:

- **Jetson Orin NX 16GB (100 TOPS)** will run our hybrid model on-board, in real time, without offloading to a remote server.
- **RGB Camera + RealSense D435i** will provide visual streams where each new scene becomes a distinct continual learning episode, grounding evaluation in real perceptual data.
- **L1 LiDAR + IMU** will capture terrain transitions (flat floor → carpet → stairs) that form natural sequential task boundaries.
- **ROS2-compatible Python SDK** will let our PyTorch model hook directly into the robot data pipeline via Unitree SDK2 and CycloneDDS, requiring no additional abstraction layer.

The Go2 Edu will validate HippoCortex beyond benchmark results, demonstrating an edge robotic system that learns, adapts, and remembers continuously, functioning much like biological systems do.

TIMELINE

Month	Phase
1–3	Stage 1: SWR generator and null-space projector implementation on Mamba; benchmarking on Split-CIFAR100 and ImageNet-R; paper submission
4–5	Stage 2a: Hybrid Mamba-Transformer architecture design and training on vision datasets
6–7	Stage 2b: Simulator validation in Mujoco / Isaac Sim Go2 environment; sequential task curriculum
8–9	Stage 2c: Go2 Edu deployment at WSO2 Colombo; real-world sequential task experiments
10	Stage 2d: Analysis, second paper submission, open-source release

REFERENCES

[1] J. Kirkpatrick et al., “Overcoming catastrophic forgetting in neural networks,” *Proc. Natl. Acad. Sci.*, vol. 114, no. 13, pp. 3521–3526, 2017.

[2] M. De Lange et al., “A continual learning survey: Defying forgetting in classification tasks,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 7, pp. 3366–3385, 2022.

[3] H. Shin, J. K. Lee, J. Kim, and J. Kim, “Continual learning with deep generative replay,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017.

[4] G. Buzsáki, “Hippocampal sharp wave-ripple: A cognitive biomarker for episodic memory and planning,” *Hippocampus*, vol. 25, no. 10, pp. 1073–1188, 2015.

[5] A. Gu and T. Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” *arXiv preprint arXiv:2312.00752*, 2023.

[6] K. Sohn, H. Lee, and X. Yan, “Learning structured output representation using deep conditional generative models,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2015.

[7] G. Saha, I. Garg, and K. Roy, “Gradient projection memory for continual learning,” in *Int. Conf. Learn. Represent. (ICLR)*, 2021.

[8] D. Cheng et al., “Mamba-CL: Optimizing selective state space model in null space for continual learning,” *arXiv preprint arXiv:2411.15469*, 2024.

[9] I. N. Lee, L. Mahmoodi, T. Le, and M. Harandi, “Exemplar-free continual learning for state space models,” *arXiv preprint arXiv:2505.18604*, 2025.

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